Problem H
Chotu ke Kanche

Time limit: 1 second

Chotu and his two friends have finished a game of kanche in the gali. Their three jars contain a, b and c kanche, respectively. Before going home, they want to share them equally.

Chotu offers to handle everyone's kanche himself. His friends don't trust him quite that much. Trust is important, but so is knowing where Chotu's hands are.

In one move, choose two different jars and transfer any positive integer number of kanche from one to the other. The source jar must contain enough kanche. Each transfer counts as **one move**, regardless of how many kanche are moved.

No kanche may be added or discarded. Jars may be empty.

Find the minimum number of moves needed to make all three jars contain the same number of kanche, or determine that it is impossible.

Input

The first line contains an integer T ($1 \le T \le 100000$), the number of test cases.

Each of the next T lines contains three integers a, b and c ($0 \le a, b, c \le 10^8$), the initial numbers of kanche in the jars.

Output

For each test case, print the minimum number of moves needed, or -1 if the jars cannot be made equal.

Sample Input 1	Sample Output 1
6 10 10 10 8 10 12 2 2 17 1 2 17 0 0 0 0 0 3	0 1 2 -1 0 2